



IoT Based Baby Hazardous Situation Detection System Using ESP32-Cam And Edge AI

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Introduction

This study provides a baby tracking system that allows working and busy parents to assure their babies' care and safety. To create a system for recognizing harmful circumstances for babies and sending safety notifications to parents.

Importance of protecting newborns and young children.

The safety of babies is essential and parents frequently confront difficulties keeping tabs on their infants, particularly when they are not in close vicinity. To address this worry, the "Baby Hazardous Situations Detection" project seeks to create a system that can detect potentially dangerous conditions for babies and notify parents in real time.

Cost/ budget

The project tries to design an affordable Baby safety locket Here is the cost of our project device:-

Component Name	Price
Lithium Lon Battery 600 mAh3.7V	BDT 290
ESP32-CAM WiFi+ Bluetooth Camera Module Development Board ESP32 With Camera Module Ov2640	BDT 848
LipoBattery Charger IS 3.7V 4 Port 350mAH	BDT 249
Boster	BDT 200
Total	BDT 1587

OBJECTIVES



Predicts the upcoming threat and alerts the user to prevent fires.



Remote Monitoring

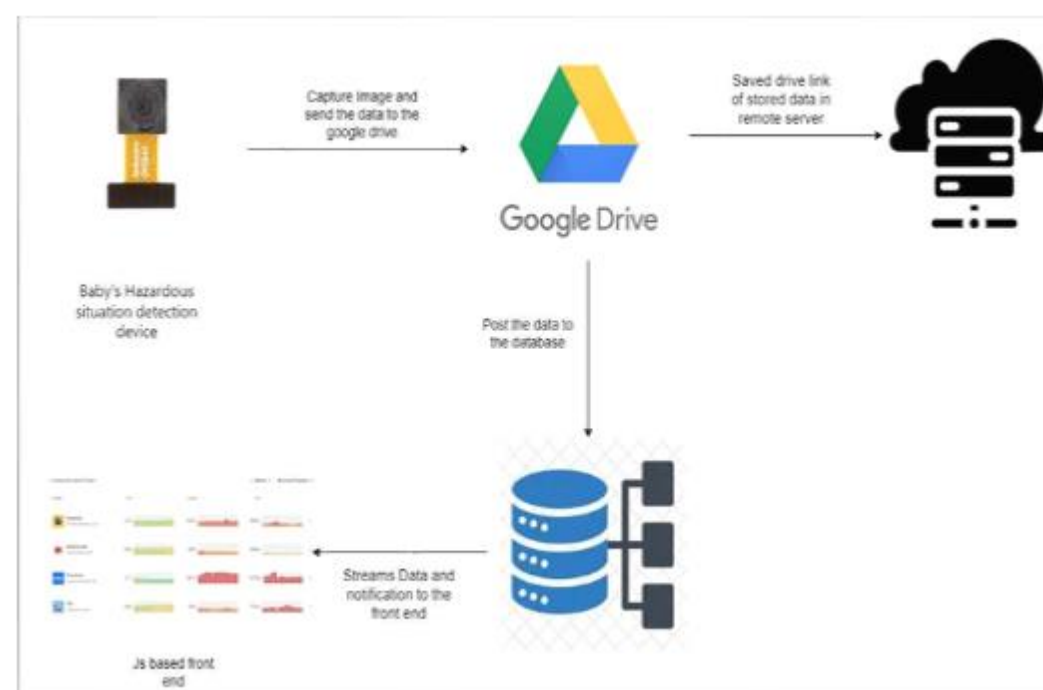


Affordable

METHODOLOGY

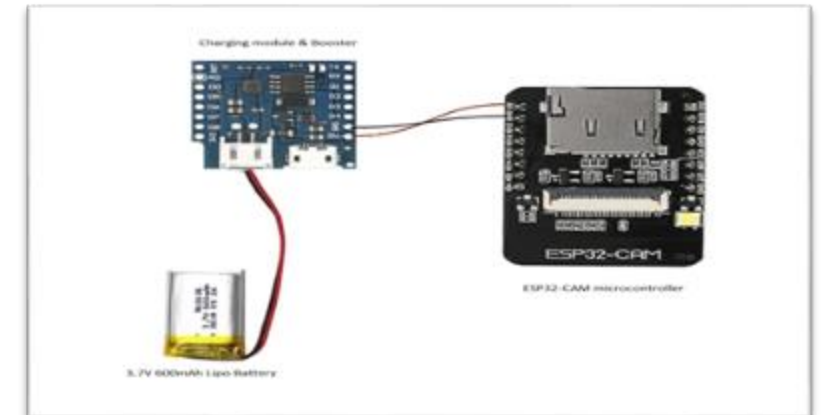
The methodology involved in building the hazardous situation detection system for babies using an ESP32-CAM microprocessor proceeded in several steps. First, a dataset comprising 300 sample images each of hazardous objects like fire, knives, and scissors was collected. This dataset was then augmented using data augmentation techniques to increase its size for robust model training. Next, the Edge Impulse platform was utilized to develop a machine learning model capable of identifying hazardous objects from the collected dataset. After training, the model was imported into the ESP32-CAM platform as a library, and necessary modifications were made to the code to enable Wi-Fi connectivity and image transmission to a cloud storage service, such as Google Drive. The ESP32-CAM was then programmed to capture images in real-time and process them using the trained model for hazard detection. Finally, thorough testing and evaluation were conducted to assess the system's performance in accurately detecting hazardous

situations in front of a baby, ensuring its reliability and effectiveness in enhancing baby safety.



SCHEMATIC DIAGRAM

Using edge impulse we have trained our model with 300 images of situations to detect hazardous situations for a baby. Like a knife or an electric switch board which can be dangerous so we have trained our model to detect such situations near the baby if these situations are detected then our device will notify the parents



Circuit diagram

DETECTED PARAMETERS



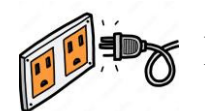
Fire



Knife



Scissor



Electric socket board



Medicine



Matchbox

CONCLUSION & FUTURE WORK

Achievements: Developed a baby locket hazardous situation detection system capable of real-time danger identification and alerting.

Future additions: Potential additions include improving the system's capabilities to detect other threats and interacting with smart devices to provide automated reactions to recognized dangers.